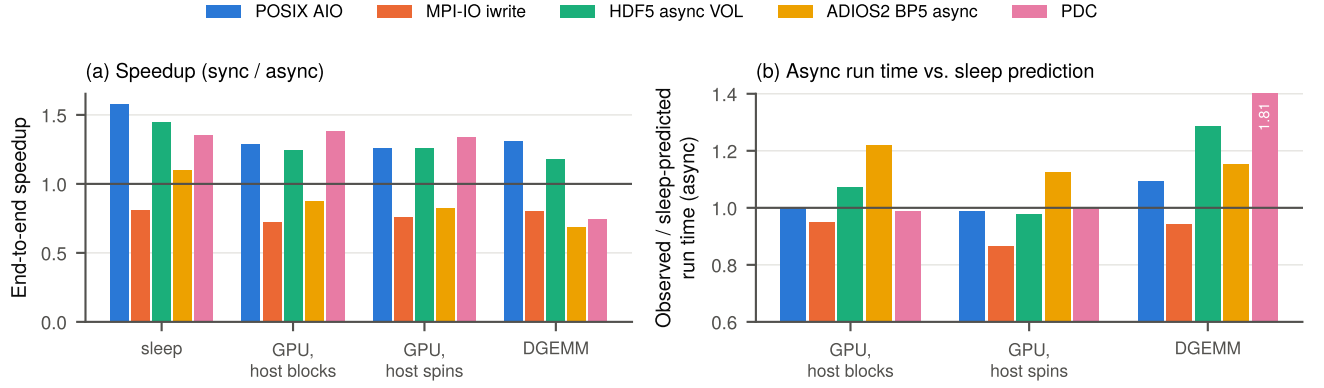
POSIX AIO
MPI-IO iwrite
HDF5 async VOL
ADIOS2 BP5 async
PDC
(a) Speedup (sync / async)
End-to-end speedup
1.5
1.0
0.5
0.0
sleep
GPU, host blocks
GPU, host spins
DGEMM
(b) Async run time vs. sleep prediction
Observed / sleep-predicted run time (async)
1.4
1.2
1.0
0.8
0.6
1.81
GPU, host blocks
GPU, host spins
DGEMM